

Coding for Computational Social Science 2026-27

v. of Sep. 21th

This course introduces students to the principles and practice of Coding, with a focus on quantitative analysis of data for Sociology and Political Sciences.

All lecture materials, exercises, and boilerplate codes are published in the course GitHub repository. Please register for a github.com/ account using your *@studenti.unimi.it* account and join <https://github.com/education>.

Online Course resources

Resource	Link
Moodle (course page, quizzes, readings)	myariel.unimi.it/course/view.php?id=12471
MS Teams (live sessions, announcements, chat)	shorturl.at/BSzTQ
GitHub (lecture materials & code)	github.com/ale66/learn-coding
Python tutor (learn coding online)	pythontutor.com/
Memory graph (learn coding online)	memory-graph.com/

Logistics

Lectures take place Thursday, 16:30-18:30, in *Aula 27* and Friday, 10:30-12:30, in *Aula 24*; both are located on the first floor of the *Conservatorio* building. A (low-fi) AV broadcast and lecture recording will be available.

Contacts

Questions on course materials, coding etc.: please post on Teams so all benefit from the answer.

Questions on personal project/studies: you may contact me on Teams anytime.

In-person office hours: Thursday after class and Friday before class.

Software Tools

Install these to follow the course and complete exercises:

- **Web browser** Firefox will be used in class.
- **Package manager** to quickly install/update software. For Win: install chocolatey.org; For Mac: install brew.sh.
- **Python interpreter** www.python.org/

- **Code editor** code.visualstudio.com/; also available online at vscode.dev/.
- **Version control software** git-scm.com/
- **A modern Git manager**, e.g., fork.dev

Calendar (TBC)

Week	Thursday	Friday
0	Onboarding Os	Simple syntax Google Colab
1	Formal languages Version control	...
2	...	...
3	...	...
4	...	...
5	...	...
6	...	...
7	...	...
8	...	...
9	...	...

Exams

Prepare and maintain on GitHub a coding project consisting of, e.g., a replication study from the CSS scientific literature or an analysis of open user-generated or admin data, e.g., from gdeltproject.org/. The instructor will present the project challenge in class. The oral exam consists of presenting the technical side of your project and showing *ownership* of the code.

Exam dates December 17-19 in person at the *Sala seminari* on the second floor of the *Passione* building (i.e., above the *Laboratorio 2* computer lab). Exam slots are bookable via Teams. Arrangements are available for Erasmus/exchange students who need to leave Milan earlier. For 2027, January 13 and 14 (TBC) and dates in March and July.